

◆ END-TO-END FLOW

KINETIC TYPE

App Flow Document

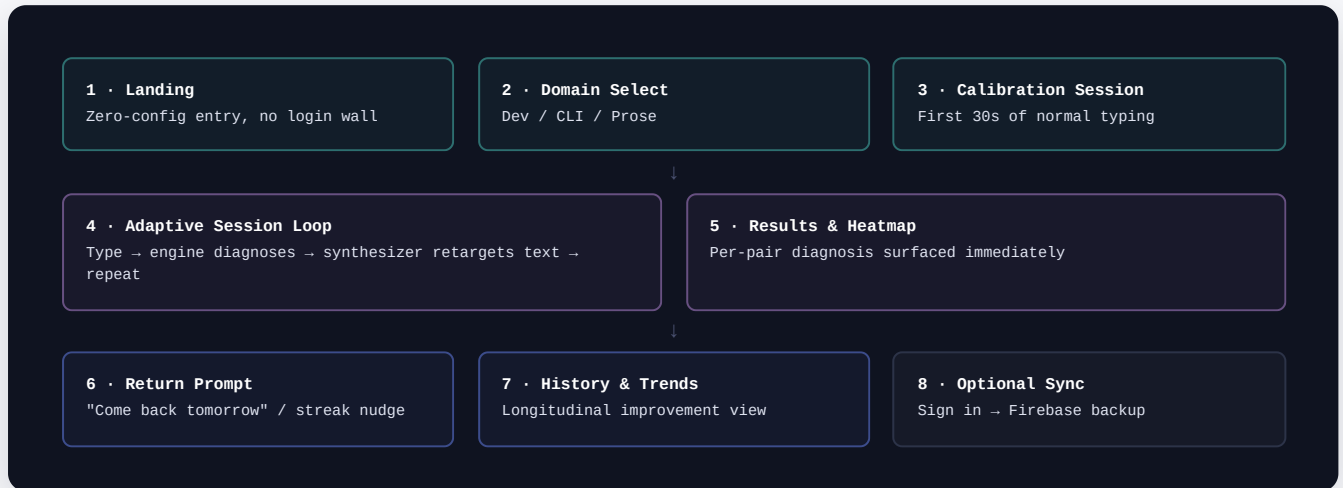
Every path a user can take through Kinetic Type — from first keystroke to multi-device sync — mapped screen by screen.

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01 Journey Overview

The entire product collapses to one repeating loop — type, diagnose, adapt, return — wrapped by a one-time onboarding and an optional identity layer.



Steps 1–5 require no account and no network. Step 8 is reachable from anywhere in the app via a persistent, non-blocking "Back up my progress" affordance — never a forced gate.

02 First-Time Onboarding Flow

STEP 1 — LANDING

User arrives with zero prior state. No modal, no tour. A single headline and the three domain cards are immediately interactive.

STEP 2 — DOMAIN SELECT

User taps Dev, CLI, or Prose. A 2-line live preview of that domain's text style renders under the card before commit, so the choice is informed, not blind.

STEP 3 — CALIBRATION (IMPLICIT)

The chosen domain's text begins streaming immediately as a normal session — there is no separate "test" framing. Internally, the first ~150 keystrokes seed the user's initial Key-Pair Performance Matrix with wide confidence intervals.

STEP 4 — FIRST DIAGNOSIS

At session end, the results screen explicitly calls out what changed: *"We found your 3 slowest transitions"* — the moment the product's core value becomes visible and concrete.

STEP 5 — SOFT BACKUP PROMPT

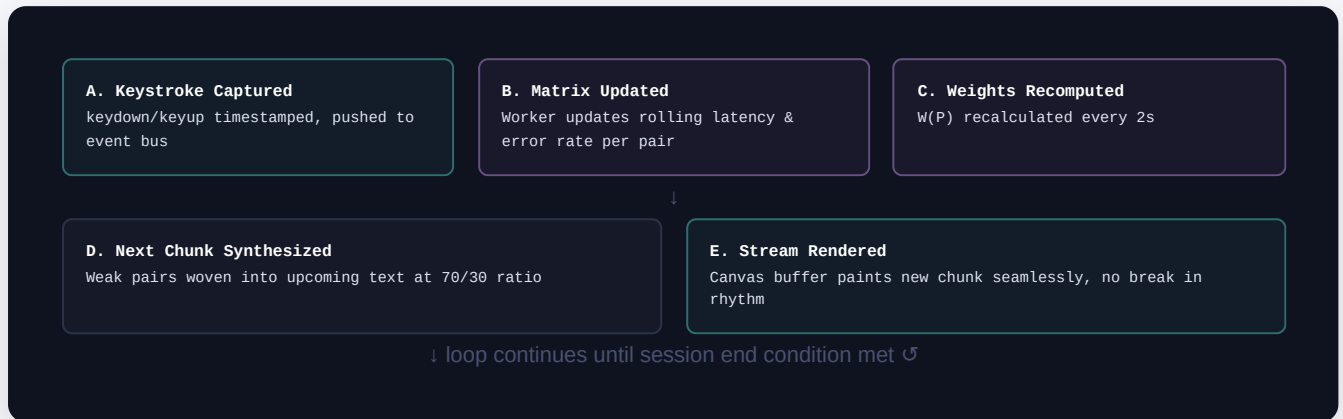
A single dismissible banner offers account backup ("so this doesn't disappear if you clear your browser"). Dismissing it never reappears mid-session, only between sessions, decaying in frequency.

DESIGN CONSTRAINT

Time from landing to first completed session must be under 60 seconds (PRD KPI). Every step above must be skippable or defaulted — the domain selector defaults to "Prose" if a user starts typing before choosing.

03 Core Session Loop

This loop repeats continuously for the duration of a session, typically executing its full cycle every few seconds.



SESSION END CONDITIONS

- Time-based limit reached (user-selected duration), or
- Character-count target reached, or
- User manually ends the session (`Esc` or explicit "Finish" control)

On any end condition, control passes to **Section 05 — Results & Analytics Flow**.

04 Adaptive Feedback Loop — Detail

This expands step D of the Core Session Loop — the Text Selection Protocol that decides exactly what the user types next.



Input: Corpus C, Latency Matrix M, Target Length K
Output: Generated Text Sequence S

1. Retrieve lowest-performing pairs `P_target` from M, ranked by $W(P)$.
2. Query domain Corpus C for tokens containing `P_target` matches.
3. Assemble S balancing 70% targeted high-weight tokens, 30% contextual flow.
4. Spaced-repetition gate re-inserts recently-mastered pairs at low density.
5. Render S into the display buffer – user never perceives a "quiz," only continuous text.

Because this protocol re-runs continuously (not once at session start), a user who suddenly fumbles a *new* transition mid-session will see it re-enter the stream within seconds — the loop is reactive, not scheduled.

USER-PERCEIVED EXPERIENCE

None of the above is visible as mechanism. The user simply experiences text that "happens to" keep testing the thing they're worst at — which is precisely the point.

05 Results & Analytics Flow

1. SESSION SUMMARY

Immediately on session end: gross/net WPM, accuracy, and the top-3 changed key pairs (improved or worsened) since the prior session.

2. HEATMAP DRILL-DOWN

User can tap into a full keyboard-shaped latency heatmap; tapping any key surfaces its specific pair breakdown and trend sparkline.

3. HISTORY & TRENDS

A rolling view plots the user's previously-flagged weak pairs over time (last 10/30/90 sessions), the primary evidence of the product working.

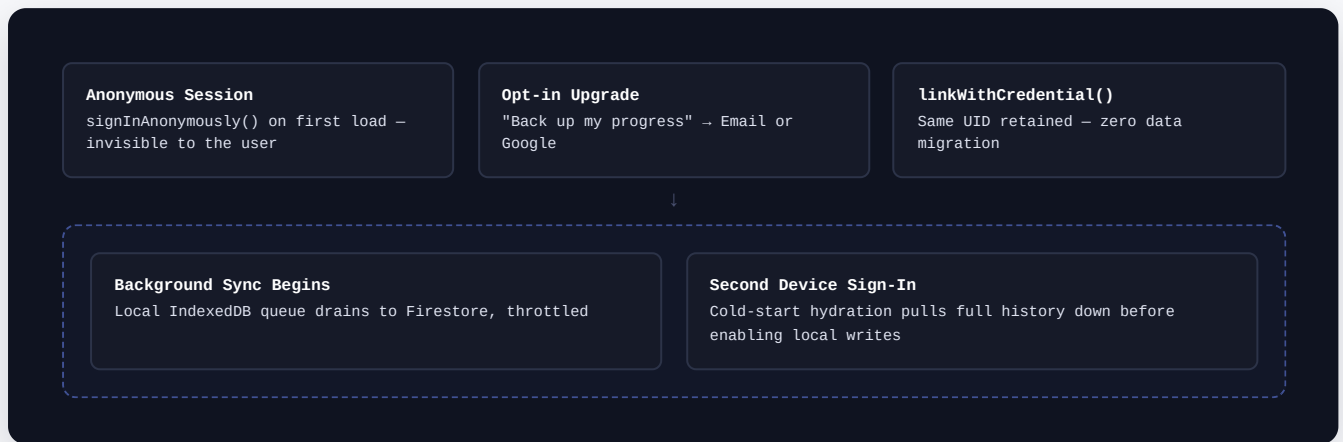
4. SHARE / EXPORT (P2)

Optional export of a session summary card as an image, or a CSV of raw session history for power users.

EXIT POINTS FROM RESULTS

- **Type Again** → returns straight to Core Session Loop (Section 03), same domain.
- **Switch Domain** → returns to Domain Select.
- **View History** → History & Trends screen.
- **Close** → Landing, with the return-prompt streak indicator now visible.

06 Account & Sync Flow



MULTI-DEVICE SCENARIO

1. User signs in on Device B with the same credential.
2. Device B pulls the complete `ngram_stats` collection once, seeding a fresh local IndexedDB store.
3. From that point, both devices write-through to Firestore; per-pair merges (weighted by occurrence count) reconcile any overlap — see TRD §06.
4. Neither device is ever blocked on the other; sync is eventually consistent, never synchronous with typing.

07 Settings & Customization

- **Session length** — time-based (1/3/5 min) or character-count target.
- **Adaptivity intensity** — default / gentle (caps targeted-word ratio below 70%).
- **Appearance** — Dark (default) / Focus (light) mode toggle.
- **Motion** — respects OS-level reduced-motion, with an explicit in-app override.
- **Data** — export history (CSV/JSON), delete local data, manage account/sync.

08 Edge Cases & Error Flows

SCENARIO	FLOW BEHAVIOR
Tab backgrounded mid-session	Timer pauses; in-flight latency samples spanning the visibility change are discarded, not recorded as false micro-pauses.
Fully offline, never signed in	No behavior change — this is the default, fully supported state.
Signed in but network drops mid-session	Sync queue simply grows locally; session itself is unaffected; a small non-blocking "offline" indicator appears.
Sync conflict (edited on two devices while one was offline)	Resolved via weighted per-pair merge on reconnect (TRD S06) — never a user-facing conflict dialog.
Browser/device lacks reliable high-res timers	App detects degraded <code>performance.now()</code> resolution and falls back to coarser session-level metrics only, disabling per-pair diagnosis with a clear in-app notice.
Local storage cleared without backup	User returns to a fresh onboarding state; no crash, no error — data loss is silent by design trade-off, mitigated by the backup prompt.

09 Screen Reference Table

SCREEN	ENTRY POINT	PRIMARY ACTIONS	EXIT POINTS
Landing	App load (cold or return visit)	Select domain, resume last domain	→ Domain Select / Session
Domain Select	Landing, "Switch Domain"	Choose Dev / CLI / Prose, preview text	→ Session
Typing Stage (Session)	Domain Select, "Type Again"	Type; pause/end session	→ Results
Results	Session end	View heatmap, type again, view history	→ Session / History / Landing
History & Trends	Results, nav menu	Filter by domain/date range, drill into a pair	→ Results / Landing
Settings	Nav menu (any screen)	Adjust session length, adaptivity, appearance, data	→ Previous screen
Sign In / Backup	Soft prompt, Settings	Email/Google sign-in, link anonymous account	→ Previous screen (sync begins in background)

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